

Methods Appendix:

Estimating Local LGBTQ Populations

July 1, 2026

1 Measurement in HPS Microdata

Let $Y_i \in \{0, 1\}$ indicate whether individual i identifies as LGBTQ. Sex is defined using sex assigned at birth, and age is inferred from year of birth and top-coded at 62.

An individual is classified as LGBTQ if sexual orientation is not straight or if gender identity is transgender. Subcategories include lesbian or gay, bisexual, queer, and transgender.

Survey weights (w_i) are retained throughout. At this stage, estimates are purely survey-based and reflect sampling variability.

2 Soft Calibration to Gallup Marginals

Household Pulse Survey (HPS) and Gallup estimates differ in survey mode, sampling variation, and field timing. Rather than imposing exact agreement, HPS-based probabilities are softly calibrated toward Gallup benchmarks.

Each individual begins with a baseline probability:

$$p_{0i} = \text{clipped}(Y_i).$$

A logit adjustment model is estimated:

$$\text{logit}(p_i) = \text{logit}(p_{0i}) + X_i\theta,$$

where X_i includes generation effects and generation-by-sex interactions.

The parameter vector θ minimizes a weighted objective combining: (i) a deviation penalty that keeps estimates close to HPS baselines, and (ii) a margin penalty that discourages deviation from Gallup targets. Optimization is performed using BFGS.

3 State-Cohort LGBT Rate Estimation

Using calibrated probabilities, state-sex-age LGBT identification rates are computed as:

$$\hat{P}(Y = 1 \mid S, \text{State}, A) = \frac{\sum_i w_i p_i^{\text{adj}}}{\sum_i w_i}.$$

Effective sample size is defined as:

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

Weekly survey waves are averaged to reduce short-term noise.

4 Age-Local Dirichlet Smoothing of Gender Composition

This step estimates the distribution of gender identity (cisgender, non-binary, transgender) within each sexual orientation category, sex assigned at birth, and exact age.

Let $g \in \{\text{cis}, \text{non-binary}, \text{trans}\}$ index gender categories. Define:

$$\hat{p}_{g,\ell,s,a} = \frac{n_{g,\ell,s,a}}{\sum_g n_{g,\ell,s,a}},$$

where $n_{g,\ell,s,a}$ denotes survey-weighted counts.

Effective sample size is:

$$n_{\text{eff},\ell,s,a} = \frac{(\sum w_i)^2}{\sum w_i^2}.$$

4.1 Age-Local Prior Construction

For a focal age a , neighboring ages $a' \neq a$ receive weights:

$$w_{a'}^{(a)} = n_{\text{eff},\ell,s,a'} \exp\left(-\frac{|a - a'|}{\tau}\right),$$

where $\tau > 0$ controls smoothness.

The age-local prior mean is:

$$m_g^{(a)} = \frac{\sum_{a' \neq a} w_{a'}^{(a)} \hat{p}_{g,\ell,s,a'}}{\sum_{a' \neq a} w_{a'}^{(a)}}.$$

4.2 Dirichlet Moment Matching

A Dirichlet prior is centered at $\mathbf{m}^{(a)}$. The concentration parameter is estimated via method of moments:

$$\alpha_0 = \text{mean}_g \left(\frac{m_g(1 - m_g)}{v_g} - 1 \right),$$

where v_g denotes weighted variance across neighboring ages. The prior vector is $\alpha_g = \alpha_0 m_g$.

4.3 Posterior Shrinkage

The smoothed estimate for focal age a is:

$$\tilde{p}_{g,\ell,s,a} = \frac{n_{\text{eff},\ell,s,a} \hat{p}_{g,\ell,s,a} + \alpha_g}{n_{\text{eff},\ell,s,a} + \alpha_0}.$$

5 Cross-State Shrinkage of LGBT Subcategory Shares

For each sex s , age bin a , and LGBT category ℓ , define:

$$\hat{\pi}_{\ell,s,a,j} = P(\ell \mid Y = 1, s, a, \text{state } j),$$

with effective sample size $n_{\text{eff},j}$.

A beta prior is estimated across states using method of moments:

$$\pi_{\ell,s,a,j} \sim \text{Beta}(\alpha, \beta).$$

Posterior shrinkage yields:

$$\tilde{\pi}_{\ell,s,a,j} = \frac{n_{\text{eff},j} \hat{\pi}_{\ell,s,a,j} + \alpha}{n_{\text{eff},j} + \alpha + \beta}.$$

Final shares are renormalized within each state–sex–age bin.

6 ACS Projection to Census Tracts

Let $N_b^{(s,a)}$ denote the ACS 5-year population count for sex $s \in \{M, F\}$ and age bin a in Census tract b . Expected LGBTQ population counts are projected to the tract level by applying state–sex–age identification rates and gender composition shares to local demographic counts.

For each combination of sex, age bin, LGBTQ subcategory ℓ , and gender identity $g \in \{m, w, nb\}$, the expected count in tract b is:

$$\hat{E}[\text{count}_{g,\ell,b}] = \sum_{s,a} N_b^{(s,a)} \cdot \hat{P}(Y = 1 \mid s, a, \text{state}) \cdot \tilde{\pi}_{\ell,s,a} \cdot \tilde{p}_{g,\ell,s,a}$$

where $\tilde{\pi}_{\ell,s,a}$ is the shrunken subcategory share from Section 5 and $\tilde{p}_{g,\ell,s,a}$ is the smoothed gender composition share from Section 4. Since sexuality and gender identity co-vary with age and sex, and these vary locally, expected LGBTQ counts vary across tracts even within a state.

Tracts that span PUMA boundaries receive population-weighted contributions from each PUMA proportional to the area of intersection.

7 PUMA Reweighting with Inverse-Distance-Weighted Smoothing

The projection in Section 6 allocates LGBTQ population based only on local age and sex composition. A second stage reweights within-state distribution using the observed geographic concentration of same-sex couples as a proxy for the broader LGBTQ population.

7.1 PUMA-Level Calibration Factors

For each PUMA p , define the uncalibrated LGBTQ share:

$$S_p = \frac{\sum_{b \in p} \hat{E}[\text{LGBTQ}_b]}{\sum_{p'} \sum_{b \in p'} \hat{E}[\text{LGBTQ}_b]}.$$

A PUMA-level signal σ_p is constructed from the observed concentration of same-sex couples among all couples, logit-transformed and mean-centered with respect to the uncalibrated LGBTQ distribution:

$$\tilde{\sigma}_p = \text{logit}\left(\frac{\text{SS couples}_p}{\text{SS couples}_p + \text{OS couples}_p}\right) - \sum_{p'} S_{p'} \text{logit}(\dots).$$

The tilt factor for PUMA p is $T_p = \exp(\gamma \tilde{\sigma}_p)$, where $\gamma > 0$ governs the strength of the reweighting. The calibrated target share is:

$$S_p^* = \frac{S_p T_p}{\sum_{p'} S_{p'} T_{p'}}.$$

The calibration factor is $\hat{c}_p = S_p^*/S_p$.

7.2 Inverse-Distance-Weighted Smoothing to Tract Level

Applying $\hat{c}_p$ uniformly to all tracts within PUMA p produces a step-function surface with hard discontinuities at PUMA boundaries. These jumps are artifacts of the PUMA delineation rather than genuine demographic transitions. To address this, calibration factors are smoothed to the tract level using inverse-distance weighting.

Let $\mathbf{x}_b$ and $\mathbf{x}_p$ denote the centroids of tract b and PUMA p , respectively. The IDW-smoothed calibration factor for tract b is:

$$\tilde{c}_b = \frac{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2} \hat{c}_p}{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2}},$$

where distances are computed in projected coordinates (EPSG:3857). Tracts within a single PUMA receive a factor close to $\hat{c}_p$; tracts near PUMA boundaries receive a blend of the neighboring factors, producing a spatially continuous surface.

All subcategory estimates in tract b are multiplied by $\tilde{c}_b$. Final estimates are rescaled so state-

level totals equal the uncalibrated totals:

$$\text{scale} = \frac{\sum_b \hat{E}[\text{LGBTQ}_b]}{\sum_b \tilde{c}_b \hat{E}[\text{LGBTQ}_b]}.$$

8 Capacity Constraint

In high-concentration areas, PUMA reweighting and state-level rescaling can push the estimated LGBTQ count for a gender group above that group's total population. A capacity constraint ensures no tract's estimates violate this bound.

For gender group $g \in \{m, w, \text{nb}\}$ and tract b , let $\hat{N}_{g,b}^{\text{LGBTQ}}$ denote the post-calibration LGBTQ count and $N_{g,b}$ the gender-group population. Define:

$$\kappa_{g,b} = \min\left(1, \frac{N_{g,b}}{\hat{N}_{g,b}^{\text{LGBTQ}}}\right).$$

All subcategory counts for gender group g in tract b are multiplied by $\kappa_{g,b}$. This preserves the relative composition of subcategories (lesbian/gay, bisexual, queer, transgender) within each gender group while ensuring the total cannot exceed the population.